ECE 332: Engineering Electromagnetics II

Lecture 17

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Quick Recap

Quick Recap (Continued)

Radiation from Large-Aperture Antennas

An **ideal wire antenna**, i.e. current source antenna, generates electromagnetic waves using currents **along a line in space** whereas an **aperture antenna** generates electromagnetic waves using **an electric field distributed across its aperture**.

Example: A Horn Antenna connected to a Coaxial Transmission Line

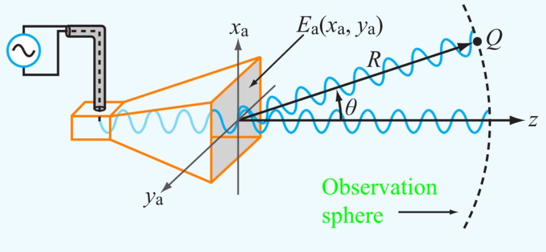


Figure 9-20 A horn antenna with aperture field distribution $E_a(x_a, y_a)$.

The horn acts as a **waveguide** that is fed by the **monopole antenna** consisting of the **protruding end of the transmission line**.

Radiation from Large-Aperture Antennas (Continued)

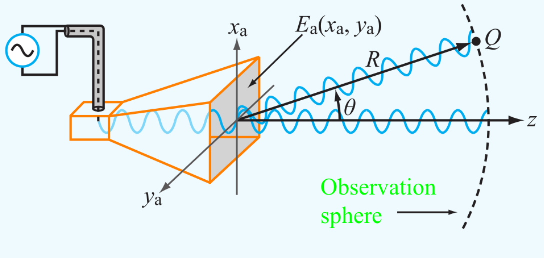


Figure 9-20 A horn antenna with aperture field distribution $E_a(x_a, y_a)$.

- **Geometry** determines the electromagnetic fields **within the horn**.
- **The aperture is the boundary** between
  - the region where the wave is guided
  - and the unbounded medium.
- **Each point** on the aperture can be thought of as an **isotropic antenna** radiating spherical waves.

Radiation from Large-Aperture Antennas (Continued)

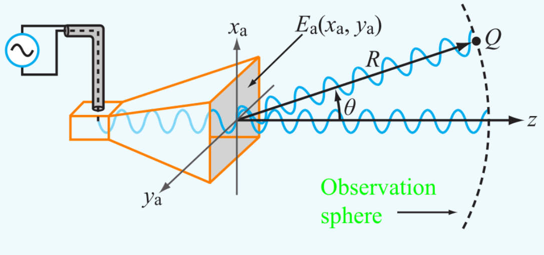


Figure 9-20 A horn antenna with aperture field distribution $E_a(x_a, y_a)$.

- The waves from each point on the aperture are **superposed** to determined the **overall wave emitted**.
- There are **two approaches** to analyzing this situation.
  - a **scalar theory** associated with Kirchhoff,
  - a **vector theory** derived from Maxwell's equations.

Radiation from Large-Aperture Antennas (Continued)

We'll explore the **scalar theory**, which is appropriate for antennas that have apertures with principal dimensions that are **at least several wavelengths long**. (Therefore, these are "large-aperture" antennas.)

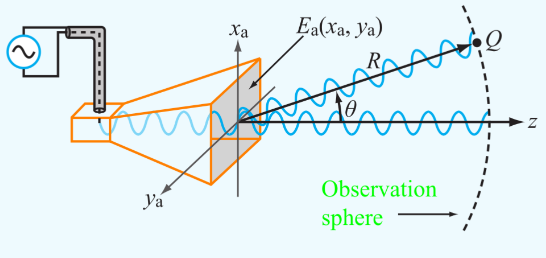


Figure 9-20 A horn antenna with aperture field distribution $E_a(x_a, y_a)$.

One application is microwave communication systems where

- high directivity is important
- $1\,\text{GHz} \le f \le 30\,\text{GHz} \Longleftrightarrow 30\,\text{cm} \ge \lambda \ge 1\,\text{cm}$

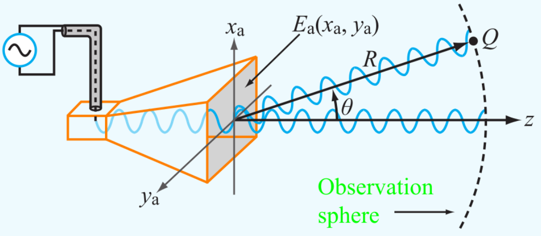


Figure 9-20 A horn antenna with aperture field distribution $E_a(x_a, y_a)$.

Define a **2D coordinate system** (x_a, y_a) with its origin at the center of the aperture. The (scalar) **electric field** $E_a(x_a, y_a)$ **at the aperture** determines the radiation that the antenna emits. The electric field $E_a(x_a, y_a)$ is called the

- **electric-field aperture distribution**
- or **illumination**.

Radiation from Large-Aperture Antennas (Continued)

More generally, assume that an antenna has a **rectangular aperture** with length ℓ_x in the x_a direction and length ℓ_y in the y_a direction.

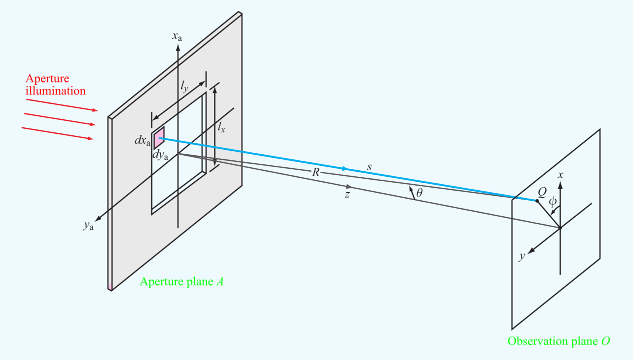


Figure 9-22 Radiation by an aperture in the x_a–y_a plane at $z = 0$.

The goal is to determine the electric field at a point Q on an "observation" plane that is parallel to the aperture.

Radiation from Large-Aperture Antennas (Continued)

For the observation plane,

- the origin lies on the z axis of the aperture,
- the z axis lies along the z axis of the aperture,
- the x and y axes are parallel to the x_a and y_a axes, respectively,
- and every point Q on the plane is in the far field of the aperture.

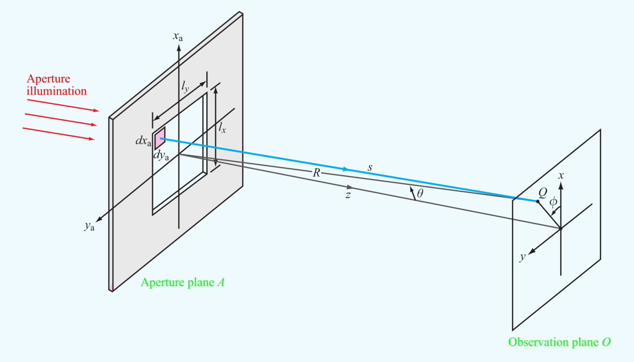


Figure 9-22 Radiation by an aperture in the x_a–y_a plane at $z = 0$.

Radiation from Large-Aperture Antennas (Continued)

Let (R, θ, ϕ) be the spherical (aperture) coordinates of a point Q on the observation plane. Here, the **elevation angle** θ is the angle from the positive z axis and ϕ is the angle from the positive x_a axis (or x axis).

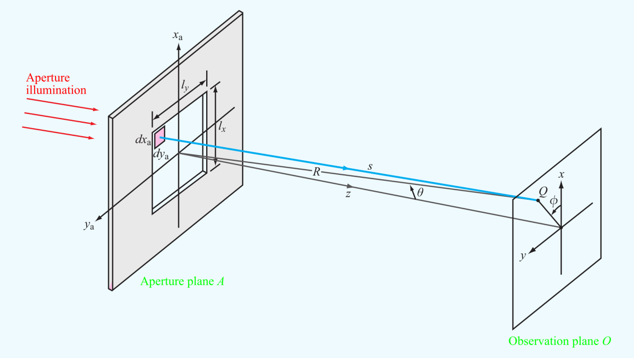


Figure 9-22 Radiation by an aperture in the x_a–y_a plane at $z = 0$.

Note: These coordinate systems differ from the coordinate system used to analyze current source antennas. (Careful when comparing formulas.)

Radiation from Large-Aperture Antennas (Continued)

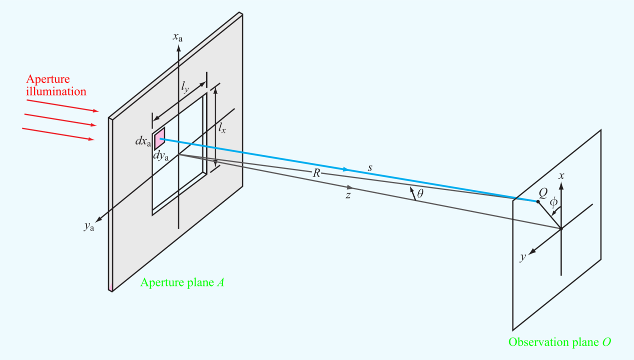


Figure 9-22 Radiation by an aperture in the x_a–y_a plane at $z = 0$.

For Q to be in the far field of the aperture,

$$R \geq 2d^2/\lambda, \tag{1}$$

where d is the **longest linear dimension** of the aperture.

Radiation from Large-Aperture Antennas (Continued)

Let $\tilde{E}(R, \theta, \phi)$ be the **phasor of the electric field at** Q and $\tilde{E}_a(x_a, y_a)$ be the **phasor of the illumination**. Then

$$\tilde{E}(R, \theta, \phi) = \frac{j}{\lambda} \left(\frac{e^{-jkR}}{R} \right) \tilde{h}(\theta, \phi), \qquad (2)$$

where

$$\tilde{h}(\theta, \phi) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \tilde{E}_a(x_a, y_a) e^{jk \sin\theta (x_a \cos\phi + y_a \sin\phi)} \, dx_a dy_a \qquad (3)$$

is the **form factor** and e^{-jkR}/R is the **spherical propagation factor**.

- The integration is done over the aperture (despite the infinite limits).
- The form factor adds the contributions from the isotropic sources across the aperture.
- The spherical propagation factor represents the attenuation of the electric field as a function of R.

Radiation from Large-Aperture Antennas (Continued)

Since the unbounded medium is lossless, the power density at point Q is

$$S(R, \theta, \phi) = \frac{|\tilde{E}(R, \theta, \phi)|^2}{2\eta_0} = \frac{|\tilde{h}(\theta, \phi)|^2}{2\eta_0 \lambda^2 R^2}. \tag{4}$$

Though not justified, the polarization of $\tilde{E}(R, \theta, \phi)$ is the same as the polarization of $\tilde{E}_a(x_a, y_a)$.

Rectangular Aperture with Uniform Illumination

Suppose that the **illumination is uniform**. This means

$$\tilde{E}_a(x_a, y_a) = \begin{cases} E_0, & -\ell_x/2 \le x_a \le \ell_x/2, \quad -\ell_y/2 \le y_a \le \ell_y/2 \\ 0, & \text{otherwise} \end{cases}, \qquad (5)$$

where E_0 is a **constant**. Also, suppose $\phi = 0$ so Q lies on the $+x$ axis.

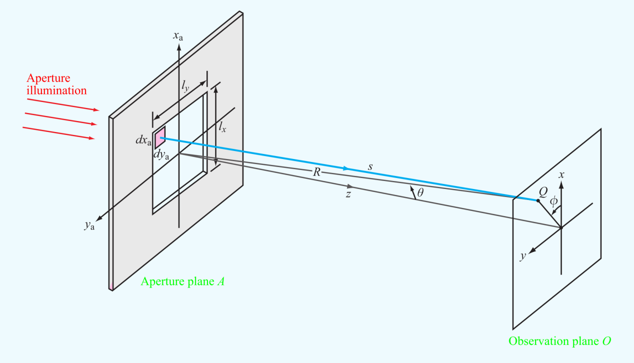


Figure 9-22 Radiation by an aperture in the x_a–y_a plane at $z = 0$.

Rectangular Aperture with Uniform Illumination (Cont.)

Under these circumstances,

$$\tilde{h}(\theta,\phi) = \tilde{h}(\theta) = \int_{-\ell_y/2}^{\ell_y/2}\int_{-\ell_x/2}^{\ell_x/2} E_0 e^{jk\sin\theta(x_a\cos(0)+y_a\sin(0))}\,dx_a dy_a$$
$$= E_0\int_{-\ell_y/2}^{\ell_y/2}\int_{-\ell_x/2}^{\ell_x/2} e^{jkx_a\sin\theta}\,dx_a dy_a. \tag{6}$$

Recall that the wavenumber $k = 2\pi/\lambda$ and define the constant

$$u = k\sin\theta = \frac{2\pi}{\lambda}\sin\theta. \tag{7}$$

Then

$$\tilde{h}(\theta) = E_0\int_{-\ell_y/2}^{\ell_y/2}\int_{-\ell_x/2}^{\ell_x/2} e^{jux_a}\,dx_a dy_a = \frac{2E_0\ell_y}{u}\sin(u\ell_x/2)$$
$$= E_0 A_p\frac{\sin(\pi\ell_x\sin\theta/\lambda)}{\pi\ell_x\sin\theta/\lambda} = E_0 A_p \mathrm{sinc}(\pi\ell_x\sin\theta/\lambda). \tag{8}$$

Rectangular Aperture with Uniform Illumination (Cont.)

Here,

$$A_p = \ell_x \ell_y \tag{9}$$

is the area of the aperture and the sinc function is defined as

$$\mathrm{sinc}(t) = \sin(t)/t. \tag{10}$$

Altogether, the **phasor of the electric field at** Q is

$$\tilde{E}(R,\theta,\phi) = \frac{j}{\lambda}\left(\frac{e^{-jkR}}{R}\right)\tilde{h}(\theta) = \frac{jE_0 A_p}{\lambda}\left(\frac{e^{-jkR}}{R}\right)\mathrm{sinc}(\pi\ell_x \sin\theta/\lambda) \tag{11}$$

and the **power density at** Q is

$$S(R,\theta) = S_0\,\mathrm{sinc}^2(\pi\ell_x \sin\theta/\lambda), \tag{12}$$

where

$$S_0 = S(R,0) = \frac{E_0^2 A_p^2}{2\eta_0 \lambda^2 R^2} \tag{13}$$

is the maximum power density.

Antenna Pattern

The **normalized radiation intensity** is

$$F(\theta) = \frac{S(R,\theta)}{S_{\text{max}}} = \frac{S(R,\theta)}{S_0}$$
$$= \text{sinc}^2(\pi \ell_x \sin\theta/\lambda)$$
$$= \text{sinc}^2(\pi\gamma), \tag{14}$$

where

$$\gamma = \frac{\ell_x}{\lambda} \sin\theta. \tag{15}$$

Note that γ is **not** the propagation constant.

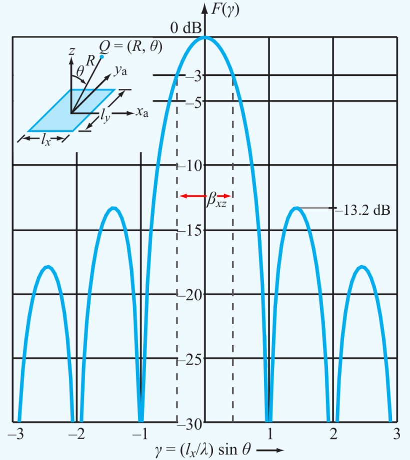


Figure 9-23 Normalized radiation pattern of a uniformly illuminated rectangular aperture in the x–z plane ($\phi = 0$).

Beamwidth

The beamwidth can be found using the normalized radiation intensity $F(\theta)$. The **half-power beamwidth in the x-z plane** is

$$\beta_{xz} \approx 0.88 \frac{\lambda}{\ell_x} \text{ (rad)} \qquad (16)$$

and the **half-power beamwidth in the y-z plane** is

$$\beta_{yz} \approx 0.88 \frac{\lambda}{\ell_y}. \qquad (17)$$

The **uniform illumination** yields the **narrowest beamwidth**.

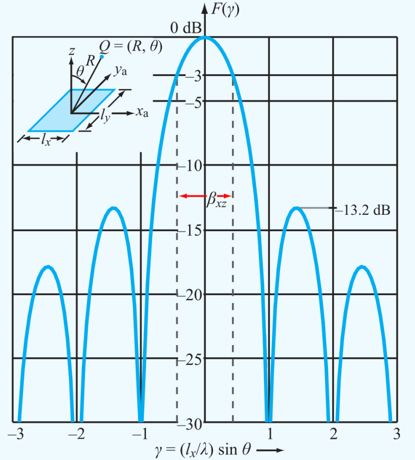


Figure 9-23 Normalized radiation pattern of a uniformly illuminated rectangular aperture in the x–z plane ($\phi = 0$).

Rectangular Aperture with Uniform Illumination (Cont.)

Applying a **taper**, which decreases $E_a(x_a, y_a)$ away from the center of the aperture, will decrease the side lobes at the expense of widening the main lobe. (This is like applying a window in signal processing.)

Directivity

The **directivity** of the antenna is

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \approx \frac{4\pi \ell_x \ell_y}{\lambda^2} = \frac{4\pi A_p}{\lambda^2}. \tag{18}$$

The **effective area** A_e **of an antenna** is defined using its directivity, specifically

$$A_e = \frac{\lambda^2}{4\pi} D. \tag{19}$$

So, for this type of antenna,

$$A_p \approx A_e. \tag{20}$$

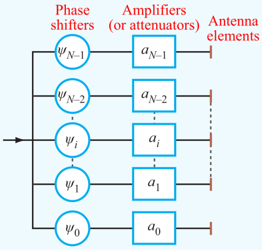
Antenna Arrays

When **two or more antennas are combined**, the direction and shape of the **overall antenna pattern can be controlled**, even when the individual antennas are very simple. In this case, the group of antennas is called an **antenna array** and operates as a single antenna. The antenna pattern of an array is controlled through the **amplitudes and relative phases of signals emitted** through the antennas. Some applications of this are

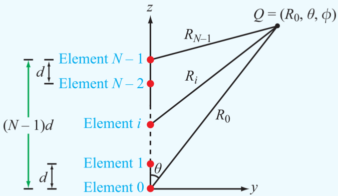
- electronic steering
- multiple-beam generation
- beam forming

Consider a **linear array of N identical transmitting antennas distributed along the z-axis**.

- All of the antennas are **fed by a single oscillator** through a branching network.
- The **phase** and **amplitude** of the signal fed to each antenna is set by a **phase shifter** and an **amplifier** in the branch for the antenna, respectively.
- An antenna is an **element** of the array.
- Antenna/Element 0 is at the **origin**.
- Antenna/Element i is **located at** $(x_i, y_i, z_i) = (0, 0, id)$ for a **constant distance** d and $i = 0, \cdots, N-1$.



(a) Array elements with individual amplitude and phase control



(b) Array geometry relative to observation point

The **phasor of the electric field intensity** in the far field for each element of the array has the form

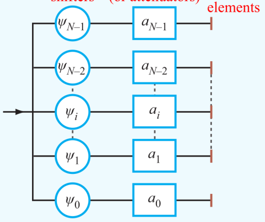
$$\tilde{E}_e(R,\theta,\phi) = \frac{e^{-jkR}}{R}\tilde{f}_e(\theta,\phi), \qquad (21)$$

where

- the **spherical propagation factor** is

$$e^{-jkR}/R \qquad (22)$$

- and $\tilde{f}_e(\theta,\phi)$ describes the **dependence of $\tilde{E}_e$ on direction**.



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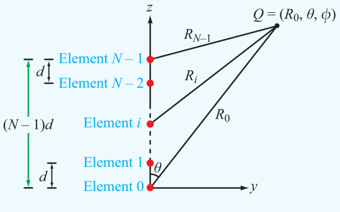
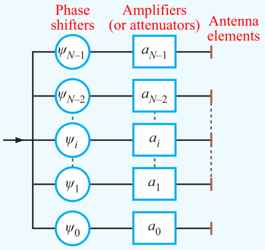


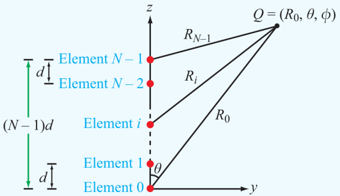
(b) Array geometry relative to observation point

Assuming that the array is transmitting into a medium with the **EM properties of free space**, such as air, the **average power density** due to $\tilde{E}_e$ is

$$S_e(R,\theta,\phi) = \frac{1}{2\eta_0}|\tilde{E}_e(R,\theta,\phi)|^2$$
$$= \frac{1}{2\eta_0 R^2}|\tilde{f}_e(\theta,\phi)|^2. \qquad (23)$$



(a) Array elements with individual amplitude and phase control



(b) Array geometry relative to observation point

Now, suppose $Q = Q(R_0, \theta, \phi)$ is a point in the **far field of all of the antennas** and that R_i is the distance from antenna i to Q. Then the **phasor of the electric field due to antenna i at Q** is

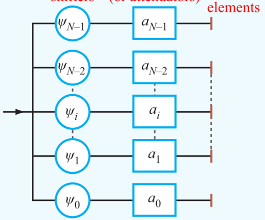
$$\tilde{E}_i(R_i, \theta, \phi) = A_i \frac{e^{-jkR_i}}{R_i} \tilde{f}_e(\theta, \phi), \qquad (24)$$

where

$$A_i = a_i e^{j\psi_i} \qquad (25)$$

is the **complex feeding coefficient** for antenna i. Here,

- the i^{th} phase shifter sets ψ_i
- and the i^{th} amplifier sets a_i.



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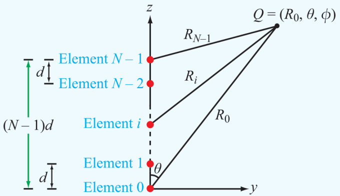


(b) Array geometry relative to observation point

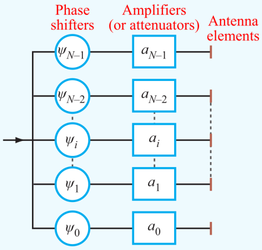
The **length of the array** is

$$\ell = (N-1)d. \tag{26}$$

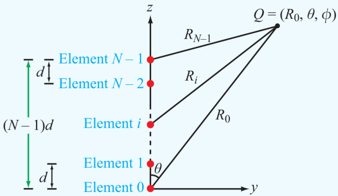
The requirement

$$R_0 \geq \frac{2\ell^2}{\lambda} = \frac{2(N-1)^2 d^2}{\lambda} \tag{27}$$

guarantees that $Q = Q(R_0, \theta, \phi)$ is **in the far field for all of the antennas**.



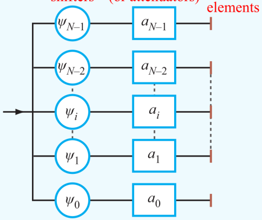
(a) Array elements with individual amplitude and phase control



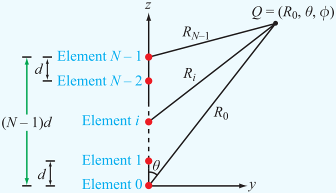
(b) Array geometry relative to observation point

The **total electric field phasor at** Q is

$$\tilde{E}(R_0,\theta,\phi) = \sum_{i=0}^{N-1} \tilde{E}_i(R_i,\theta,\phi)$$
$$= \sum_{i=0}^{N-1} A_i \frac{e^{-jkR_i}}{R_i} \tilde{f}_e(\theta,\phi)$$
$$= \left(\sum_{i=0}^{N-1} A_i \frac{e^{-jkR_i}}{R_i} \right) \tilde{f}_e(\theta,\phi). \qquad (28)$$



(a) Array elements with individual amplitude and phase control



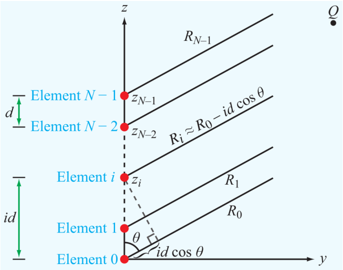
(b) Array geometry relative to observation point

Linear Antenna Arrays (Continued)

Since Q is in the far field of all of the antennas, it's possible to make 2 approximations in the expression for $\tilde{E}(R_0, \theta, \phi)$.

1. For R_i in the denominators of the spherical propagation factors,

$$R_i \approx R_0 \tag{29}$$



2. Assume that the line segments connecting Antenna i to Q are parallel for all i. This leads to the approximation

$$e^{-jkR_i} \approx e^{-jk(R_0 - z_i)} = e^{-jk(R_0 - id\cos\theta)} = e^{-jkR_0} e^{jkid\cos\theta}. \tag{30}$$

Applying these approximations,

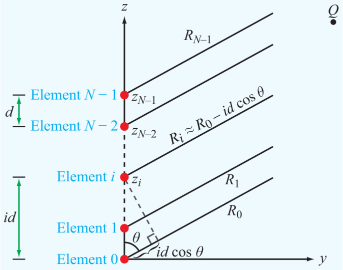
$$\tilde{E}(R_0, \theta, \phi) = \left(\sum_{i=0}^{N-1} A_i \frac{e^{-jkR_i}}{R_i} \right) \tilde{f}_e(\theta, \phi) = \tilde{f}_e(\theta, \phi) \left(\frac{e^{-jkR_0}}{R_0} \right) \sum_{i=0}^{N-1} A_i e^{jkid\cos\theta}. \tag{31}$$

Corresponding to the phasor of the electric field

$$\tilde{E}(R_0,\theta,\phi) = \tilde{f}_e(\theta,\phi)\left(\frac{e^{-jkR_0}}{R_0}\right)\sum_{i=0}^{N-1} A_i e^{jkid\cos\theta}, \tag{32}$$

the average power density of the array is

$$\begin{aligned} S(R_0,\theta,\phi) &= \frac{1}{2\eta_0}|\tilde{E}(R_0,\theta,\phi)|^2 \\ &= \frac{|\tilde{f}_e(\theta,\phi)|^2}{2\eta_0 R_0^2}\left|\sum_{i=0}^{N-1} A_i e^{jkid\cos\theta}\right|^2 \\ &= S_e(R_0,\theta,\phi)F_a(\theta). \end{aligned} \tag{33}$$



Here,

$$S_e(R_0,\theta,\phi) = \frac{|\tilde{f}_e(\theta,\phi)|^2}{2\eta_0 R_0^2} \tag{34}$$

is the **average power density due to an individual element**, as before, and

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} A_i e^{jkid\cos\theta} \right|^2 \tag{35}$$

is the **array factor**, which is the far-field radiation intensity of the array with isotropic radiators in place of the antennas.

The equation

$$S(R_0, \theta, \phi) = S_e(R_0, \theta, \phi) F_a(\theta) \tag{36}$$

implies that the overall power density can be determined by multiplying

- the far-field power pattern $F_a(\theta)$ for a corresponding array of isotropic radiators
- and the power density $S_e(R_0, \theta, \phi)$ for an individual antenna of the type used in the array.

This is called the **pattern multiplication principle**.

Linear Antenna Arrays (Continued)

Using the polar forms for the feeding coefficients, the array factor can be written as

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} a_i e^{j\psi_i} e^{jkid\cos\theta} \right|^2 . \tag{37}$$

Therefore, the array factor (and, hence, the antenna pattern) is determined by the **array amplitude distribution**

$$a_0, a_1, \cdots, a_{N-1} \tag{38}$$

and the **array phase distribution**

$$\psi_0, \psi_1, \cdots, \psi_{N-1}. \tag{39}$$

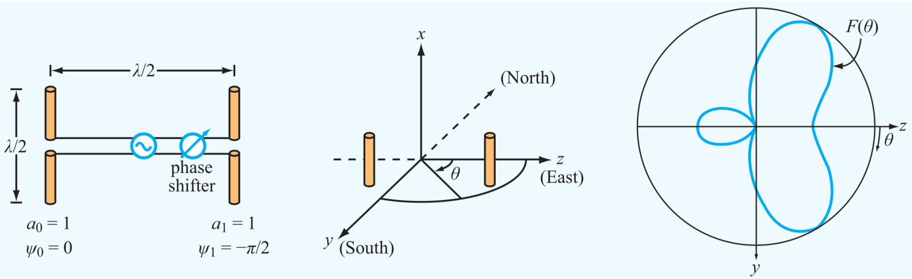
More specifically,

- the *array amplitude distribution* controls the **shape**
- and the *array phase distribution* controls the **direction**

of the array antenna pattern.

Array of Two Half-Wave Dipoles

Consider the array of two half-wave dipole antennas intended for transmitting EM waves with wavelength λ placed a distance $\lambda/2$ apart.



(a) Dipole array (b) Observation plane (c) Horizontal-plane pattern

The z-axis passes through the centers of the antennas at a height $\lambda/4$ above the ground. (The ground is the plane $x = -\lambda/4$.) Both antennas are oriented along the x-direction. One antenna is located at $z = -\lambda/4$ and the other one is located at $z = \lambda/4$. (Different than assumed before.)

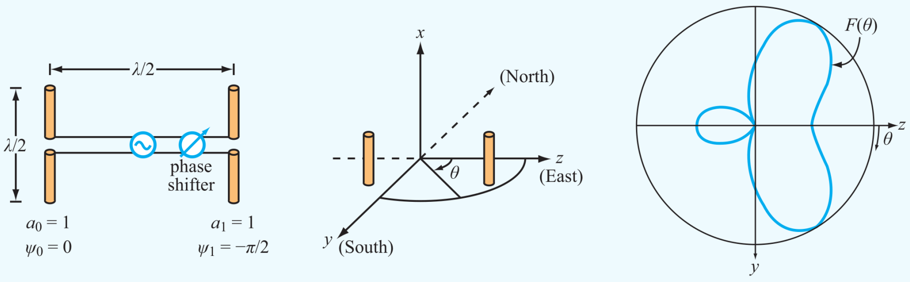
Array of Two Half-Wave Dipoles (Continued)

The **amplitudes of the feeding coefficients** are

$$a_0 = a_1 = 1 \tag{40}$$

and there is a phase shift of $-\pi/2$ radians between the emitted signals so the **phases of the feeding coefficients** are

$$\psi_0 = 0 \quad \text{and} \quad \psi_1 = -\pi/2. \tag{41}$$



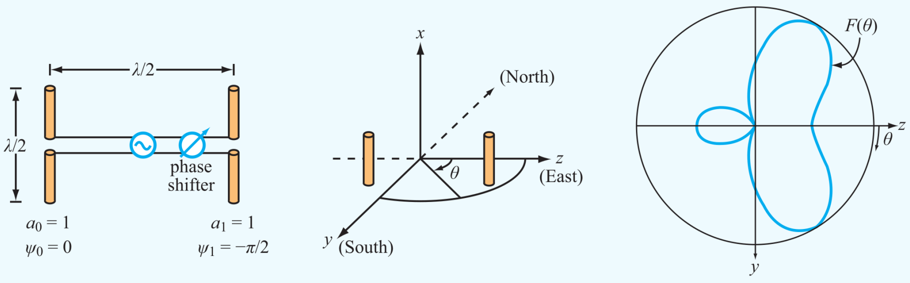
(a) Dipole array (b) Observation plane (c) Horizontal-plane pattern

Array of Two Half-Wave Dipoles (Continued)

Setting $\psi_0 = 0$ and $\psi_1 = -\pi/2$ amounts to measuring the phase relative to the phase of Antenna 0. **Phases are usually measured**

- relative to the phase of one antenna
- or relative to the phase of the source.

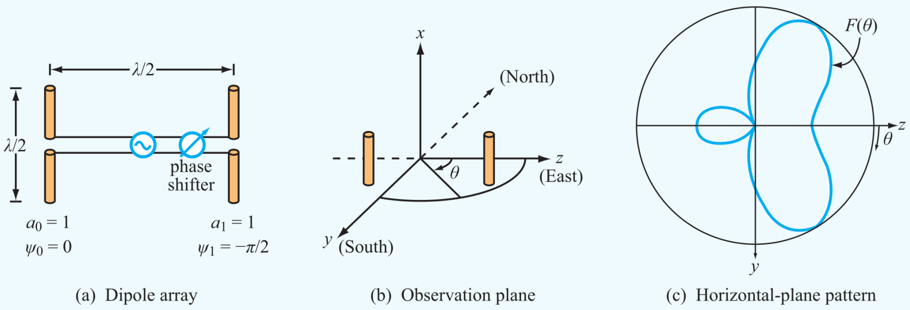
Notice that the dipoles are **oriented differently** than in the previous analysis. In the current configuration, the angle θ sweeps **a plane perpendicular to the axes of the dipoles**, like ϕ previously.



(a) Dipole array (b) Observation plane (c) Horizontal-plane pattern

Array of Two Half-Wave Dipoles (Continued)

So, in a horizontal plane, parallel to the y-z plane, an individual half-wave dipole radiates uniformly at a value that only depends on the specific plane and R and doesn't depend on θ.



(a) Dipole array (b) Observation plane (c) Horizontal-plane pattern

We'll consider the y-z plane, specifically. In this plane, the **average power density of an individual half-wave dipole** is

$$S_e = S_0 = \frac{15 I_0^2}{\pi R^2} \tag{42}$$

and

$$S(R,\theta) = S_0 F_a(\theta). \tag{43}$$

In this case,

$$k = 2\pi/\lambda, \quad d = \lambda/2 \quad \Longrightarrow \quad kd = \pi, \tag{44}$$

and the **array factor** is

$$\begin{aligned} F_a(\theta) &= \left| \sum_{i=0}^{1} a_i e^{j\psi_i} e^{jkid\cos\theta} \right|^2 = \left| a_0 e^{j\psi_0} e^{j0} + a_1 e^{j\psi_1} e^{j\pi\cos\theta} \right|^2 \\ &= \left| 1 + e^{-j\pi/2} e^{j\pi\cos\theta} \right|^2 = \left| 1 + e^{j(\pi\cos\theta - \pi/2)} \right|^2. \end{aligned} \tag{45}$$

It's possible to express this in a more familiar form. For any value x,

$$\begin{aligned} 1 + e^{jx} &= e^{jx/2}\left(e^{-jx/2} + e^{jx/2}\right) = 2e^{jx/2}\left(\frac{e^{jx/2} + e^{-jx/2}}{2}\right) \\ &= 2e^{jx/2}\cos(x/2). \end{aligned} \tag{46}$$

Array of Two Half-Wave Dipoles (Continued)

Let $x = \pi \cos\theta - \pi/2$. Then, the array factor is

$$F_a(\theta) = \left|1 + e^{jx}\right|^2 = \left|2e^{jx/2}\cos(x/2)\right|^2 = 4\left|e^{jx/2}\right|^2 |\cos(x/2)|^2$$
$$= 4\cos^2(x/2) = 4\cos^2\left(\frac{\pi}{2}\cos\theta - \frac{\pi}{4}\right). \tag{47}$$

Altogether, the **average power density of the array** in the y-z plane is

$$S(R,\theta) = 4S_0\cos^2\left(\frac{\pi}{2}\cos\theta - \frac{\pi}{4}\right). \tag{48}$$

The function $\cos^2\alpha$ has maxima when $\cos\alpha = \pm 1$, i.e. when $\alpha = m\pi$ for any integer m. So, if there is a solution to the equation

$$m\pi = \frac{\pi}{2}\cos\theta - \frac{\pi}{4} \tag{49}$$

for some integer m, the power density reaches a maximum value of $4S_0$.

Array of Two Half-Wave Dipoles (Continued)

Rearranging this equation and multiplying it by $2/\pi$ implies

$$\cos\theta = 2m + \frac{1}{2}. \tag{50}$$

For $m \neq 0$, $|2m + 1/2| > 1$. So, the only possible solution occurs when $m = 0$ and

$$\cos\theta = \frac{1}{2} \implies \theta = \frac{\pi}{3} + 2\pi n, \frac{5\pi}{3} + 2\pi p \tag{51}$$

for any integers n and p. In the current situation, the angles satisfying $0 \leq \theta < 2\pi$ cover the entire plane. Within this range, the **array factor reaches a maximum value of 4** when

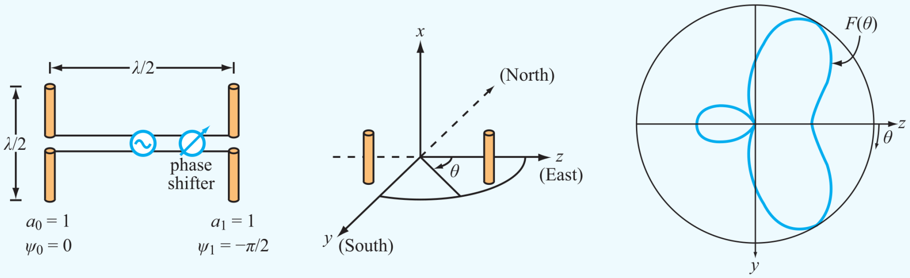
$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}. \tag{52}$$

So, in the y-z plane, the **maximum power density of the array** is

$$S_{\max} = 4S_0 \tag{53}$$

and the **normalized radiation intensity of the array** is

$$F(\theta) = \frac{S(R,\theta)}{S_{\max}} = \cos^2\left(\frac{\pi}{2}\cos\theta - \frac{\pi}{4}\right). \tag{54}$$



(a) Dipole array (b) Observation plane (c) Horizontal-plane pattern

Module 9.5 on the book's website allows you to plot the antenna pattern for a two-dipole array with different values for the array parameters and Example 9-6 considers the inverse problem of finding the parameters of the array that result in a specific array antenna pattern.

N-Element Array with Equal Phase

Consider the **original N-element array** along the z-axis with **equal spacing** d between neighboring elements and **equal phases**, i.e.

$$\psi_i = \psi_0 \quad \text{for all } i. \tag{55}$$

This is called a **broadside array**. In this case, the **array factor** is

$$\begin{aligned} F_a(\theta) &= \left| \sum_{i=0}^{N-1} a_i e^{j\psi_i} e^{jkid\cos\theta} \right|^2 = \left| \sum_{i=0}^{N-1} a_i e^{j\psi_0} e^{jkid\cos\theta} \right|^2 \\ &= \left| e^{j\psi_0} \sum_{i=0}^{N-1} a_i e^{jkid\cos\theta} \right|^2 = \left| e^{j\psi_0} \right|^2 \left| \sum_{i=0}^{N-1} a_i e^{jkid\cos\theta} \right|^2 \\ &= \left| \sum_{i=0}^{N-1} a_i e^{jkid\cos\theta} \right|^2, \end{aligned} \tag{56}$$

i.e.

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} a_i e^{jkid\cos\theta} \right|^2. \tag{57}$$

So, the **phase difference between neighboring elements** is entirely due to the distance between them and the direction θ and is

$$\gamma = kd\cos\theta = \frac{2\pi d}{\lambda}\cos\theta. \tag{58}$$

(This γ again has no relationship to the propagation constant.) With this definition, the **array factor** is

$$F_a(\gamma) = \left| \sum_{i=0}^{N-1} a_i e^{ji\gamma} \right|^2. \tag{59}$$

Now, suppose that **all of the amplitudes are the same**, as well, i.e.

$$a_i = a \in \mathbb{R} \quad \text{for all } i \text{ with } a > 0. \tag{60}$$

N-Element Array with Equal Phase (Continued)

With equal phases and equal amplitudes, the array factor is

$$F_a(\gamma) = a^2 \left| \sum_{i=0}^{N-1} e^{ji\gamma} \right|^2 = a^2 \left| 1 + e^{j\gamma} + e^{j2\gamma} + \cdots + e^{j(N-1)\gamma} \right|^2. \quad (61)$$

Let

$$f_a(\gamma) = 1 + e^{j\gamma} + e^{j2\gamma} + \cdots + e^{j(N-1)\gamma} \quad (62)$$

so that

$$F_a(\gamma) = a^2 \left| f_a(\gamma) \right|^2. \quad (63)$$

Here,

$$f_a(\gamma) = \left(e^{j\gamma}\right)^0 + \left(e^{j\gamma}\right)^1 + \left(e^{j\gamma}\right)^2 + \cdots + \left(e^{j\gamma}\right)^{N-1}, \quad (64)$$

which is a finite geometric series of the form $\sum_{n=0}^{N-1} r^n$ with $r = e^{j\gamma}$. As shown in the book (or any calculus textbook),

$$\sum_{n=0}^{N-1} r^n = \frac{1 - r^N}{1 - r} \quad (65)$$

N-Element Array with Equal Phase (Continued)

and

$$f_a(\gamma) = \frac{1 - e^{jN\gamma}}{1 - e^{j\gamma}} = \frac{e^{jN\gamma/2}}{e^{j\gamma/2}} \left(\frac{e^{-jN\gamma/2} - e^{jN\gamma/2}}{e^{-j\gamma/2} - e^{j\gamma/2}} \right)$$

$$= e^{j(N-1)\gamma/2} \frac{\sin(N\gamma/2)}{\sin(\gamma/2)}. \quad (66)$$

Therefore,

$$F_a(\gamma) = a^2 \left| f_a(\gamma) \right|^2 = a^2 \left| e^{j(N-1)\gamma/2} \frac{\sin(N\gamma/2)}{\sin(\gamma/2)} \right|^2$$

$$= a^2 \left| e^{j(N-1)\gamma/2} \right|^2 \left| \frac{\sin(N\gamma/2)}{\sin(\gamma/2)} \right|^2$$

$$= a^2 \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}, \quad (67)$$

N-Element Array with Equal Phase (Continued)

i.e. **when the amplitudes and phases are equal** the array factor is

$$F_a(\gamma) = a^2 \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}. \tag{68}$$

Notice that

$$\begin{aligned} |f_a(\gamma)| &= |1 + e^{j\gamma} + e^{j2\gamma} + \cdots + e^{j(N-1)\gamma}| \\ &\leq |1| + |e^{j\gamma}| + |e^{j2\gamma}| + \cdots + |e^{j(N-1)\gamma}| \\ &= N. \end{aligned} \tag{69}$$

When

$$kd\cos\theta = \gamma = 0, \quad \text{i.e. } \theta = \frac{\pi}{2}, \tag{70}$$

$|f_a(\gamma)|$ has this maximum value and

$$F_a(\gamma) = F_a(0) = a^2|f_a(0)|^2 = a^2N^2 = F_{a,\max}. \tag{71}$$

N-Element Array with Equal Phase (Continued)

Therefore, the **normalized array factor** is

$$F_{an} = \frac{F_a(\gamma)}{F_{a,\max}} = \frac{1}{a^2 N^2}\left(a^2 \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}\right)$$
$$= \frac{\sin^2(N\gamma/2)}{N^2 \sin^2(\gamma/2)}$$
$$= \frac{\sin^2\left(\frac{N\pi d}{\lambda}\cos\theta\right)}{N^2 \sin^2\left(\frac{\pi d}{\lambda}\cos\theta\right)}. \qquad (72)$$

This could be the same as the antenna pattern but isn't always since the power density has an additional factor of $S_e(R_0, \theta, \phi)$, which is the power density of an individual element.

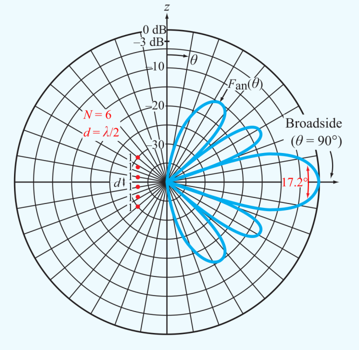


Figure 9-30 Normalized array pattern of a uniformly excited six-element array with interelement spacing $d = \lambda/2$.

Homework

Read Chapter 9.

Homework 4 is posted and due Friday, June 5 at 10 pm. I might add more problems from Chapter 9 to the assignment.

Exam 2 will be on Tuesday, June 9, from 1:30 – 3:20 pm.